



MAX9295A

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CSI-2 to GMSL2/GMSL1 Serializer

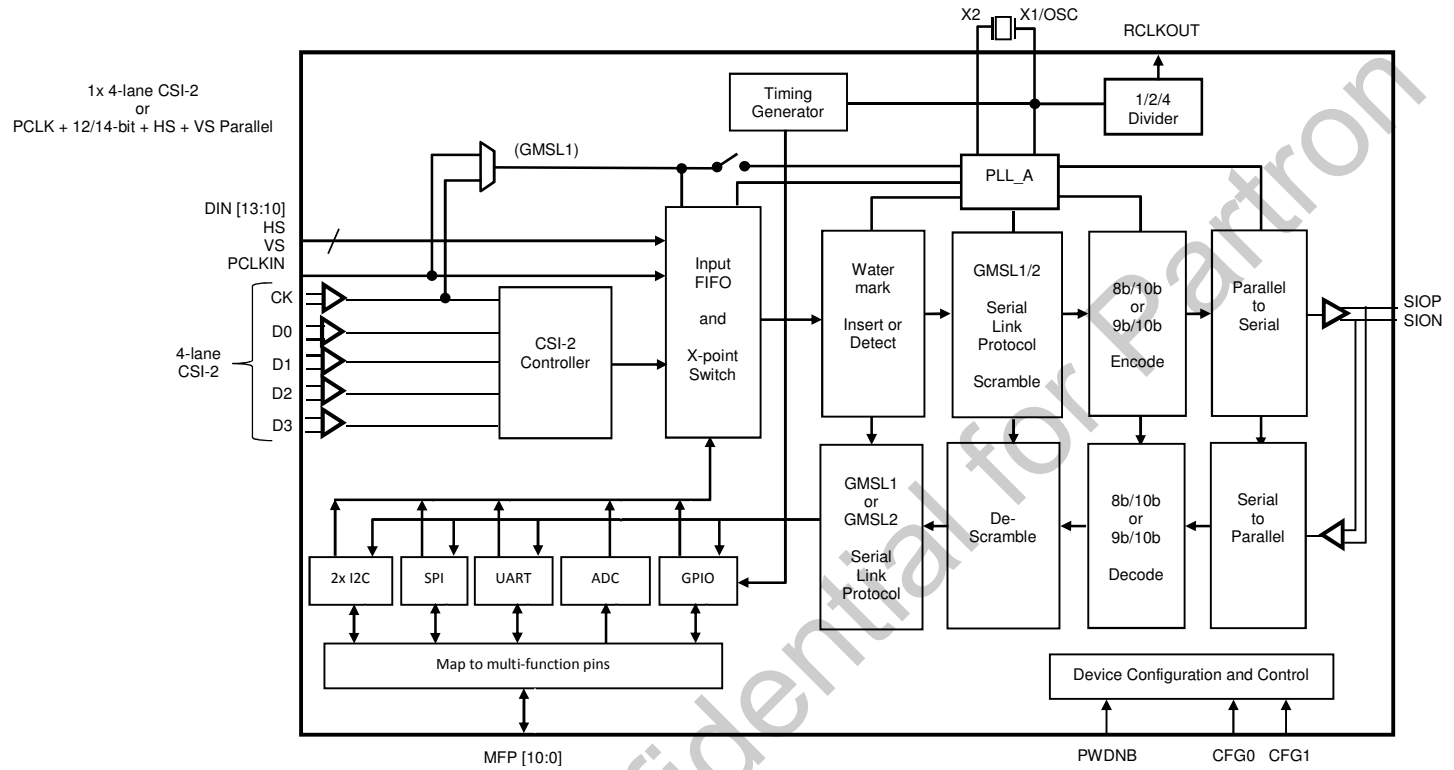
General Description

The MAX9295A converts MIPI CSI-2 or LVCMOS parallel to GMSL2 or GMSL1 while sending and receiving control channel data, enabling transmission of video and bi-directional control data over cables up to 15 meters. GMSL2 operates at fixed rate of 3Gbps or 6Gbps in the forward direction and 187.5Mbps in the reverse direction. In GMSL1 mode, the MAX9295A can be paired with first-generation 3.12Gbps or 1.5Gbps GMSL1 deserializers, or operate up to 4.5Gbps with GMSL2 deserializers in GMSL1 mode. Operation is specified over the automotive temperature range of -40°C to +105°C and the device is AEC-Q100 Grade 2 qualified. Cable can be coax or shielded twisted pair (STP).

Features

- 4-lane MIPI CSI-2 v1.3 port
- MIPI D-PHY v1.2 at 80Mbps - 2.5Gbps per lane
- 150MHz clock parallel input (12/14-bit data + H + V)
- Line fault monitor
- Supports RAW8/10/12/14/16/20, RGB565/666/888, YUV422 8/10-bit, User Defined and Generic Long Packet data types
- 16-channel CSI-2 virtual channel support
- Selectable GMSL2 or backward-compatible GMSL1 mode
- Operates up to 4.5Gbps with GMSL2 deserializers operating in GMSL1 mode
- CSI-2 ECC and Checksum error detection and flagging
- Watermark generator and checker
- XTAL oscillator or reference input with divided or undivided reference output
- On-chip timing generator for video sync signal generation
- High speed (187.5Mbps) backchannel for reduced-size power-over-cable (POC) LC filter
- Multi-function pins (MFP) programmable as GPO sync output for surround view camera systems
- Multi-function pins (MFP) programmable as I2C, SPI, UART, sampled or register-programmed GPIO
- 4 hardware-selectable device addresses
- 3 Internal and 3 external supply voltage monitor (10-bit ADC)
- Works with 50Ω coax cable or 100Ω STP
- Crosspoint switch remaps input to any desired deserializer output order
- CSI-2 lane polarity flip
- CSI-2 lane reassignment
- Eye opening monitor (EOM) on reverse channel for real-time jitter margin measurement
- Remote/local wake-up from sleep mode
- Software reset
- Concurrent control channel transactions with settable priority level
- Programmable spread spectrum on serial I/O for EMI reduction
- Forward and reverse channel PRBS for BER testing of serial link
- 32-bit CRC protection of video data
- 16-bit CRC protection of control channel data
- Retransmission of control channel data upon CRC error detection
- 5mm x 5mm TQFN or TQFN side-wettable package with 0.5mm pitch
- 1.7V to 1.9V analog supply
- Core supply accepts +/-5% tolerance supply with nominal value of 1.3V, 1.2V, or 1.0V
- 1.7V to 3.6V I/O supply
- -40°C to +105°C operating temperature
- Meets AEC-Q100 Grade 2 specification

MAX9295A Block Diagram



Absolute Maximum Ratings

VDD18 to Ground	-0.3V to +2.0V
VDD to Ground	-0.3V to +1.9V
VDDIO to Ground	-0.3V to +3.9V
SIOP, SION to Ground	+0.5V to +2.0V
LMN ₁ to Ground	-0.3V to +2.0V
D-PHY pins to Ground	-0.3V to +TBD
All Other Pins to Ground	-0.3V to +(V _{VDDIO} + 0.3V) ¹
Continuous Power Dissipation, T _A = +70°C	
Multi-Layer Board	
TQFN-SW and TQFN (derate 34.5mW/°C above +70°C)	2758mW
Single-Layer Board	
TQFN-SW and TQFN (derate 21.3mW/°C above +70°C)	1702mW
Junction Temperature	+150°C
Storage Temperature Range	-40°C to +150°C
Soldering temperature (reflow)	+260°C

The ground is internally bonded to the package's exposed pad (EP). The EP must be connected to the PCB ground plane.

¹for VDDIO ≤ +3.6V

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Package Information

32-pin TQFN-SW (Side-wettable)

Package Code	T3255Y+8
Package Outline Document Number	21-100156
Land Pattern Document Number	90-0013
Single-Layer Board	
Junction-to-Ambient Thermal Resistance (θ _{JA})	47°C/W
Junction-to-Case Thermal Resistance (θ _{JC})	1.7°C/W
Four-Layer Board	
Junction-to-Ambient Thermal Resistance (θ _{JA})	29°C/W
Junction-to-Case Thermal Resistance (θ _{JC})	1.7°C/W

32-pin TQFN

Package Code	T3255+8
Package Outline Document Number	21-0140
Land Pattern Document Number	90-0013
Single-Layer Board	
Junction-to-Ambient Thermal Resistance (θ _{JA})	47°C/W
Junction-to-Case Thermal Resistance (θ _{JC})	1.7°C/W
Four-Layer Board	
Junction-to-Ambient Thermal Resistance (θ _{JA})	29°C/W
Junction-to-Case Thermal Resistance (θ _{JC})	1.7°C/W

Ordering Information

PART NUMBER	TEMP RANGE	PIN-PACKAGE
TBD	-40°C to +105°C	32 TQFN-EP*
TBD	-40°C to +105°C	32 TQFN-EP*
TBD	-40°C to +105°C	32 TQFN-SW-EP**
TBD	-40°C to +105°C	32 TQFN-SW-EP**

/V denotes an automotive qualified product.

+Denotes a lead(Pb)-free/RoHS-compliant package.

T = Tape and reel.

*EP = Exposed pad.

**EP = Exposed pad side-wettable flanked package.

DC ELECTRICAL CHARACTERISTICS

($V_{DD18} = 1.7V$ to $1.9V$, $V_{DD} = 0.95V$ to $1.05V$ or $V_{DD} = 1.14V$ to $1.37V$, $V_{DDIO} = 1.7V$ to $3.6V$, $T_A = -40^{\circ}C$ to $+105^{\circ}C$, Maximum Junction Temperature = $125^{\circ}C$, EP connected to PCB ground, typical values are at, $V_{DD18} = V_{DDIO} = 1.8V$, $V_{DD} = 1.0V$, $T_A = 25^{\circ}C$, unless otherwise noted.) (Note 1)

GMSL2 Forward Channel Serial I/O (SIOP, SION)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Output Voltage Swing (Single-ended)	V_O	$R_L = 100\Omega \pm 1\%$	300	400	500	mV		
Output Voltage Swing (Differential)	V_{OD}	$R_L = 100\Omega \pm 1\%$	600	800	1000	mV		
Change in VOD between Complimentary Output States	ΔV_{OD}	$R_L = 100\Omega \pm 1\%$			25	mV		
Output Offset Voltage	V_{OS}	$R_L = 100\Omega \pm 1\%$	$V_{DD18} - 0.55$	$V_{DD18} - 0.45$	$V_{DD18} - 0.35$	V		
Change in VOS between Complimentary Output States	ΔV_{OS}	$R_L = 100\Omega \pm 1\%$			25	mV		
Output Termination Resistance (Internal)	R_O	From SIOP or SION to V_{DD18}	50	55	60	Ω		

GMSL1 Forward Channel Serial I/O (SIOP, SION)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Output Voltage Swing (Single-ended)	V_O	$R_L = 100\Omega \pm 1\%$, coax mode	375	500	625	mV		
Output Voltage Swing (Differential)	V_{OD}	$R_L = 100\Omega \pm 1\%$, TP mode	600	800	1000	mV		
Change in VOD between Complimentary Output States	ΔV_{OD}	$R_L = 100\Omega \pm 1\%$			25	mV		
Output Offset Voltage	V_{OS}	$R_L = 100\Omega \pm 1\%$, coax mode	$V_{DD18} - 0.65$	$V_{DD18} - 0.55$	$V_{DD18} - 0.45$	V		
		$R_L = 100\Omega \pm 1\%$, TP mode	$V_{DD18} - 0.55$	$V_{DD18} - 0.45$	$V_{DD18} - 0.35$	V		
Change in VOS between Complimentary Output States	ΔV_{OS}	$R_L = 100\Omega \pm 1\%$			25	mV		
Output Termination Resistance (Internal)	R_O	From SIOP or SION to V_{DD18}	50	55	60	Ω		

GMSL1 Reverse Control Channel Receiver (SIOP, SION)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
High Switching Threshold	V_{CHR}	HIM Disabled			27	mV		
		HIM Enabled			40	mV		
Low Switching Threshold	V_{CLR}	HIM Disabled	-27			mV		
		HIM Enabled	-40			mV		

I/O (GPIO_, MOSI, MISO)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
High-Level Input Voltage	V_{IH}		$0.7 \times V_{DDIO}$			V		

Low-Level Input Voltage	V_{IL}				$0.3 \times V_{DDIO}$	V		
High-Level Output Voltage	V_{OH}	$I_{OH} = -4mA$	$V_{DDIO} - 0.4$			V		
Low-Level Output Voltage	V_{OL}	$I_{OL} = 4mA$			0.4	V		
Input Current	I_{IN}	$V_{IN} = 0$ to V_{DDIO} All pull-up/pull-down devices disabled.			1	μA		
Input Capacitance	C_{IN}	Each pin (note 2)		5		pF		
Input pull-up/ pull-down Resistance	R_{IN}	40k Ω pull-up or pull-down enabled		40		k Ω		
		1M Ω pull-up or pull-down enabled		1		M Ω		

Open-Drain I/O (SDA, SCL, GPI, ODO)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
High-Level Input Voltage	V_{IH2}		$0.7 \times V_{DDIO}$			V		
Low-Level Input Voltage	V_{IL2}				$0.3 \times V_{DDIO}$	V		
Low-Level Open-Drain Output Voltage	V_{OL}	$I_{OL} = 4mA$			0.4	V		
Input Current	I_{IN}	$V_{IN} = 0$ to V_{DDIO} All pull-up/pull-down devices disabled.			1	μA		
Input Capacitance	C_{IN}	Each pin (note 2)		5		pF		
Input pull-up/ pull-down Resistance	R_{IN}	40k Ω pull-up or pull-down enabled		40		k Ω		
		1M Ω pull-up or pull-down enabled		1		M Ω		

Open-Drain Outputs (LOCK, ERRB, LFLTB)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Low-Level Open-Drain Output Voltage	V_{OL}	$I_{OL} = 4mA$			0.4	V		
Input pull-up/ pull-down Resistance	R_{IN}	40k Ω pull-up or pull-down enabled		40		k Ω		
		1M Ω pull-up or pull-down enabled		1		M Ω		

Power-Down Input (PWDNB)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
High-Level Input Voltage	V_{IH}		$0.7 \times V_{DDIO}$			V		
Low-Level Input Voltage	V_{IL}				$0.3 \times V_{DDIO}$	V		
Input Current	I_{IN}	$V_{IN} = 0$ to V_{DDIO}			5	μA		
Input Capacitance	C_{IN}	(note 2)		5		pF		
Internal Pull-down Resistor	R_{PD}			1		M Ω		

Inputs (PCLK, D0-D13, HS, VS, RX, CNTL, RO)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
High-Level Input Voltage	V_{IH}		$0.7 \times V_{DDIO}$			V		
Low-Level Input Voltage	V_{IL}				$0.3 \times V_{DDIO}$	V		
Input Current	I_{IN}	$V_{IN} = 0$ to V_{DDIO} All pull-up/pull-down devices disabled.			1	μA		

Input Capacitance	C_{IN}	Each pin (note 2)		5		pF		
Input pull-up/ pull-down Resistance	R_{IN}	40k Ω pull-up or pull-down enabled		40		k Ω		
		1M Ω pull-up or pull-down enabled		1		M Ω		

Push-Pull Outputs (TX, GPO, RCLKOUT, DPLL_OUT, SS1, SS2, BNE, SCLK)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
High-Level Output Voltage	V_{OH}	$I_{OH} = -4mA$	$V_{DDIO} - 0.4$			V		
Low-Level Output Voltage	V_{OL}	$I_{OL} = 4mA$			0.4	V		
Input pull-up/ pull-down Resistance	R_{IN}	40k Ω pull-up or pull-down enabled		40		k Ω		
		1M Ω pull-up or pull-down enabled		1		M Ω		

LINE FAULT DETECTION INPUT (LMN0, LMN1)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Short-to-Battery Range	I_{SB0}	Current measured out of LMN0			-25	μA		
Open Range	I_{O0}	Current measured out of LMN0	-15		6.5	μA		
Normal Range	I_{N0}	Current measured out of LMN0	9.5		16	μA		
Short-to-Ground Range	I_{SG0}	Current measured out of LMN0	20			μA		
Open Pin Voltage	V_{O0}	Pin LMN0		1.25		V		
Short-to-Battery Range	I_{SB1}	Current measured out of LMN1			-12.5	μA		
Open Range	I_{O1}	Current measured out of LMN1	-7.5		2	μA		
Line-to-Line Short Range	I_{LLS1}	Current measured out of LMN1	4		5	μA		
Normal Range	I_{N1}	Current measured out of LMN1	7		8	μA		
Short-to-Ground Range	I_{SG1}	Current measured out of LMN1	10			μA		
Open Pin Voltage	V_{O1}	Pin LMN1		0.75		V		

Reference Clock Input (Crystal) (X1, X2)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
X1 Input Capacitance	C_{IN-X1}			5.5		pF		
X2 Input Capacitance	C_{IN-X2}			4		pF		
X2 Limit Resistor	R_{LIM}			1.2		k Ω		
Feedback Resistor	R_{FB}			10		k Ω		
Transconductance	g_m			28		mA/V		

Reference Clock Input (External Input on X1; X2 floating)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
High-Level Input Voltage	V_{IH}	$V_{DDIO} = 1.7V$ to $3.6V$	0.9			V		
Low-Level Input Voltage	V_{IL}	$V_{DDIO} = 1.7V$ to $3.6V$			+0.4	V		
Input Impedance	R_{IN}	Note 2		10		k Ω		
X1 Input Capacitance	C_{IN-X1}	Note 2		5.5		pF		

Power Supply

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Supply Current	I _{DD1}	GMSL2 mode, 6Gbps, 4-lane CSI-2 input, 1.3Gbps per lane, Color Bar pattern		10	15	mA		
	I _{DD2}			112	125	mA		
	I _{DD3}			104	180	mA		
	I _{DD4}			106	185	mA		
	I _{DD5}	GMSL1 mode, High Bandwidth Mode, 4.5Gbps, 4-lane CSI-2 input, 900Mbps per lane, Color Bar pattern		TBD	TBD	mA		
	I _{DD6}			TBD	TBD	mA		
	I _{DD7}			TBD	TBD	mA		
	I _{DD8}			TBD	TBD	mA		
Sleep-mode Supply Current	I _{DDS}	Wake up receiver enabled			TBD	μA		
Power-down Supply Current	I _{DDZ}	PWDNB = ground			TBD	μA		

D-PHY HS Receiver

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Common-mode voltage HS receive Mode	V _{CMRX(DC)}	Figure A	70		330	mV		
Differential input high threshold	V _{IDTH}				40	mV		
Differential input low threshold	V _{IDTL}		-40			mV		
Differential input impedance	Z _{ID}		80	100	125	Ω		

D-PHY LP Receiver

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
High Level Input Voltage	V _{IH}	Figure A	740			mV		
Low Level Input Voltage	V _{IL}				550	mV		
Input hysteresis	V _{HYST}		25			mV		

ESD Protection

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
SIOP, SION	V _{ESD}	Human Body Model, R _D = 1.5kΩ, C _S = 100pF		±4		kV		
		IEC61000-4-2, R _D = 330Ω, C _S = 150pF, Contact discharge		±4				
		IEC61000-4-2, R _D = 330Ω, C _S = 150pF, Air discharge		±8				
		ISO10605, R _D = 2kΩ, C _S = 330pF, Contact discharge		±4				
		ISO10605, R _D = 2kΩ, C _S = 330pF, Air discharge		±8				
		AEC - Q100-011 Rev-C1, Charged Device Model (CDM)		500		V		
Corner Pins	V _{ESD}	Human Body Model, R _D = 1.5kΩ, C _S = 100pF		±4		kV		
		AEC - Q100-011 Rev-C1, Charged Device Model (CDM)		750		V		

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
All Other Pins	V_{ESD}	Human Body Model, $R_D = 1.5k\Omega$, $C_S = 100pF$		± 2		kV		
		AEC - Q100-011 Rev-C1, Charged Device Model (CDM)		750		V		

Monitor Analog-to-Digital Converter (ADC)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Resolution				10				
Input Voltage Range	V_{IN}	Internal reference, No divider, No internal buffer	0		V_{REF}	V		
		Internal reference, 2:1 divider, No internal buffer	0		$2 \times V_{REF}$			
		Internal reference, 4:1 divider, No internal buffer	0		3.6			
		No divider, Internal buffer	0.1		V_{REF}			
Input Resistance	R_{IN}	Input configured for 2:1 voltage division		60		k Ω		
		Input configured for 4:1 voltage division		40		k Ω		
		Buffered Input		TBD		M Ω		
Integral Nonlinearity	INL			± 2		LSB		
Differential Nonlinearity	DNL			± 1		LSB		
Offset Error				± 2		LSB		
Gain Error		Excluding offset and reference errors		± 1		LSB		
Total Unadjusted Error	TUE			± 1		LSB		
Power-Down Current				10		nA		
Active Load Current				260		μA		

AC ELECTRICAL CHARACTERISTICS

($V_{DD18} = 1.7V$ to $1.9V$, $V_{DD} = 0.95V$ to $1.05V$ or $V_{DD} = 1.14V$ to $1.37V$, $V_{DDIO} = 1.7V$ to $3.6V$, $T_A = -40^\circ C$ to $+105^\circ C$, Maximum Junction Temperature = $125^\circ C$, EP connected to PCB ground, typical values are at, $V_{DD18} = V_{DDIO} = 1.8V$, $V_{DD} = 1.0V$, $T_A = 25^\circ C$, unless otherwise noted.) (Note 1)

SWITCHING CHARACTERISTICS (note 2)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Serial Output Rise Time	t_R	20% to 80%, $V_{OD} = 800mV$ differential, $R_L = 100\Omega$, 400mV single-ended $R_L = 50\Omega$		50	70	ps		
Serial Output Fall Time	t_F	80% to 20%, $V_{OD} = 800mV$ differential, $R_L = 100\Omega$, 400mV single-ended $R_L = 50\Omega$		50	70	ps		
Total Serial Output Jitter (Differential output)	t_{TSOJ1}	PRBS7, measured at $V_{OD} = 0V$ differential. Forward channel, spread spectrum disabled.		0.25		UI (p-p)		
Deterministic Serial Output Jitter (Differential output)	t_{DSOJ2}	PRBS7, measured at $V_{OD} = 0V$ differential		0.15		UI (p-p)		
Total Serial Output Jitter (Single-ended output)	t_{TSOJ1}	PRBS7, measured at $V_O/2$		0.25		UI (p-p)		
Deterministic Serial Output Jitter (Single-ended output)	t_{DSOJ2}	PRBS7, measured at $V_O/2$		0.15		UI (p-p)		

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Serializer Latency	t_{SL}	From CSI-2 input data bit to corresponding GMSL2 output bit		TBD		T_{PCLK}		
Video Initialization Time	$t_{LNKSTRT}$	From CSI-2 clock start to transmission of GMSL2 CSI-2 data		2		ms		
Lock Time	t_{LOCK}	PWDNB high to LOCK high		7		ms		
GPI-to-GPO Delay (reverse channel)	t_{GPDR}	Delay-compensated Mode		10		μs		
		Non-delay-compensated Mode		3		μs		
GPI-to-GPO Delay (forward channel)	t_{GPDF}	Delay-compensated Mode		TBD		TBD		
		Non-delay-compensated Mode		TBD		TBD		

I2C or UART PORT TIMING

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Output Rise Time	t_R	30% to 70%, $C_L = 10pF$ to 100pF, 1k Ω pullup to VDDIO (note 2)	20		150	ns		
Output Fall Time	t_F	70% to 30%, $C_L = 10pF$ to 100pF, 1k Ω pullup to VDDIO (note 2)	20		150	ns		

I2C Timing

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
SCL Clock Frequency	f_{SCL}	Low f_{SCL} range: (I2CMST_BT = 010, I2CSLV_SH = 10)	9.6		100	kHz		
		Mid f_{SCL} range : (I2CMST_BT 101, I2CSLV_SH = 01)	100		400			
		High f_{SCL} range : (I2CMST_BT = 111, I2CSLV_SH = 00)	400		1000			
Start Condition Hold Time	$t_{HD:STA}$	f_{SCL} range, Low	4			μs		
		f_{SCL} range, Mid	0.6					
		f_{SCL} range, High	0.26					
Low Period of SCL Clock	t_{LOW}	f_{SCL} range, Low	4.7			μs		
		f_{SCL} range, Mid	1.3					
		f_{SCL} range, High	0.5					
High Period of SCL Clock	t_{HIGH}	f_{SCL} range, Low	4			μs		
		f_{SCL} range, Mid	0.6					
		f_{SCL} range, High	0.26					
Repeated Start Condition Setup Time	$t_{SU:STA}$	f_{SCL} range, Low	4.7			μs		
		f_{SCL} range, Mid	0.6					
		f_{SCL} range, High	0.26					
Data Hold Time	$t_{HD:DAT}$	f_{SCL} range, Low	0			ns		
		f_{SCL} range, Mid	0					
		f_{SCL} range, High	0					
Data Setup Time	$t_{SU:DAT}$	f_{SCL} range, Low	250			ns		
		f_{SCL} range, Mid	100					
		f_{SCL} range, High	50					
Setup Time for Stop Condition	$t_{SU:STO}$	f_{SCL} range, Low	4			us		
		f_{SCL} range, Mid	0.6					
		f_{SCL} range, High	0.26					
Bus Free Time	t_{BUF}	f_{SCL} range, Low	4.7			us		
		f_{SCL} range, Mid	1.3					
		f_{SCL} range, High	0.5					
Data Valid Time	$t_{VD:DAT}$	f_{SCL} range, Low			3.45	us		
		f_{SCL} range, Mid			0.9			
		f_{SCL} range, High			0.45			
Data Valid Acknowledge Time	$t_{VD:ACK}$	f_{SCL} range, Low			3.45	us		
		f_{SCL} range, Mid			0.9			
		f_{SCL} range, High			0.45			
Pulse width of spikes suppressed	t_{SP}	f_{SCL} range, Low			50	ns		
		f_{SCL} range, Mid			50			
		f_{SCL} range, High			50			
Capacitive load each bus line	C_B	Note 2			100	pF		

SPI Master

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Operating Frequency	f_{MCK}				50	MHz		
SCLK Period	t_{MCK}			$1/f_{MCK}$		ns		
SCLK Output Pulse-Width High/Low	t_{MCH}, t_{MCL}			$t_{MCK}/2$		ns		
MOSI Output Hold Time	t_{MOH}	After SCLK Rising Edge		$t_{MCK}/2$		ns		
MOSI Output Valid	t_{MOV}	Before SCLK Rising Edge		$t_{MCK}/2$		ns		
MISO Input Setup Time	t_{MIS}	Before SCLK Rising Edge	5			ns		
MISO Input Hold Time	t_{MIH}	After SCLK Rising Edge	1.5			ns		

SPI Slave

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Operating Frequency	f_{MCK}				50	MHz		
SCLK Period	t_{MCK}			$1/f_{MCK}$		ns		
SCLK Input Pulse-Width High/Low	t_{MCH}, t_{MCL}			$t_{MCK}/2$		ns		
MISO Output Hold Time	t_{MOH}	After SCLK Rising Edge		$t_{MCK}/2$		ns		
MISO Output Valid	t_{MOV}	Before SCLK Rising Edge		$t_{MCK}/2$		ns		
MOSI Input Setup Time	t_{MIS}	Before SCLK Rising Edge	5			ns		
MOSI Input Hold Time	t_{MIH}	After SCLK Rising Edge	1.5			ns		

Power-Down Input (PWDNB)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Low Level Assertion Time	t_{PWNB}			1		ms		

Reference Clock Input Requirements (Crystal) (X1, X2)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Frequency	f_{XTAL}			25		MHz		
Frequency Stability + Frequency Tolerance	f_{TN}		-200		+200	ppm		

Reference Clock Requirements (External Input on X1; X2 floating)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Frequency	f_{REF}			25		MHz		
Frequency Stability + Frequency Tolerance	f_{TN}		-200		+200	ppm		
Input Jitter		Forward data rate 6Gbps, Reverse data rate 187.5Mbps, Sinusoidal jitter <1MHz (rising edge)			600	ps (p-p)		
Input Duty Cycle	T_{DUTY}		40		60	%		
Input Rise Time	t_R	10% to 90%			8	ns		

Parallel Clock Input (PCLK)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Duty Cycle	DC		35	50	65	%		
Transition Time	t_R, t_F	150MHz, 20% to 80%		1		ns		
		75MHz, 20% to 80%		2		ns		

Parallel Inputs (D0-D13, HS, VS)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Data to PCLK Input Setup	t_{SET}	Figure TBD	1			ns		
Data to PCLK Input Hold	t_{HOLD}	Figure TBD	1			ns		

Reference Clock Output (RCLKOUT)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Frequency	f_{REF}	Crystal or Ref Clock Input		25		MHz		
Rise Time	t_R	20% to 80%, $C_L = 10pF$, 12.5MHz (25MHz divided by 2)			2	ns		
Fall Time	t_F	80% to 20%, $C_L = 10pF$, 12.5MHz (25MHz divided by 2)			2	ns		
Jitter	t_j	$C_L = 10pF$, Rising or falling edge, 12.5MHz (25MHz divided by 2)		TBD		ps (p-p)		

Digital PLL Output (DPLL_OUT)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Frequency	f_{DPLL}	$C_L = 10pF$	TBD		75	MHz		
Rise Time	t_R	20% to 80%, $C_L = 10pF$			2	ns		
Fall Time	t_F	80% to 20%, $C_L = 10pF$			2	ns		
Jitter Generation	t_j	Rising or Falling Edge, Deterministic jitter + 14 * Random Jitter, 27MHz			TBD	ps		

D-PHY HS Receiver

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Common-mode interference beyond 450MHz	$\Delta V_{CMRX(HF)}$	Note 4			100	mV		
		Data rate > 1.5Gbps			50	mV		
Common-mode interference beyond 50MHz - 450MHz	$\Delta V_{CMRX(LF)}$	Note 3, 6			50	mV		
		Data rate > 1.5Gbps	-25		25	mV		
Common-mode termination	C_{CM}	Note 5			60	pF		

D-PHY LP Receiver

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Input pulse rejection	θ_{SPIKE}	Figure B			300	V-ps		
Minimum pulse width response	T_{MIN-RX}		20			ns		
Peak interference amplitude	V_{INT}				200	mV		
Interference frequency	f_{INT}		450			MHz		

D-PHY Data-Clock Timing

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
UI instantaneous	U_{INST}	Figure C	0.4		12.5	ns		
Data to Clock Setup Time	$T_{SETUP(RX)}$	Figure C	0.15			U_{INST}		
Data to Clock Hold Time	$T_{HOLD(RX)}$	Figure C	0.15			U_{INST}		

D-PHY Global Operation Timing

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Timeout for receiver to detect absence of Clock transitions and disable the Clock Lane HS-RX.	$T_{CLK-MISS}$	Figure F			60	ns		
Time interval during which the HS receiver ignores any Clock Lane HS transitions, starting from the beginning of $T_{CLK-PREPARE}$.	$T_{CLK-SETTLE}$	Figure E	95		300	ns		
Time for the Clock Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$.	$T_{CLK-TERM-EN}$	Figure F	Time for Dn to reach $V_{TERM-EN}$		38	ns		
Time for the Data Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$.	$T_{D-TERM-EN}$	Figure D	Time for Dn to reach $V_{TERM-EN}$		35ns + $4 \times U_{INST}$	ns		

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Time interval during which the HS receiver ignores any Data Lane HS transitions, starting from the beginning of $T_{HS-PREPARE}$.	$T_{HS-SETTLE}$	Figure D	85ns + $6 \times UI_{INST}$		145ns + $10 \times UI_{INST}$	ns		
Time interval during which the HS-RX ignores any transitions on the Data Lane, following a HS burst.	$T_{HS-SKIP}$	Figure D	40		55ns + $4 \times UI_{INST}$	ns		

Monitor Analog-to-Digital Converter (ADC_)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Throughput				ADC_CLK/1024		ksps		
Conversion Time				430		μs		
ADC Setup Time	T_{ADC_SETUP}	Excluding V_{REF} power-up time		30		μs		
ADC Clock	F_{ADCCLK}			2.5	5	MHz		

Note 1: Limits are 100% tested at $T_A = +25^\circ\text{C}$ and $T_A = +105^\circ\text{C}$. Limits over the operating temperature range and relevant supply voltage range are guaranteed by design and characterization.

Note 2: Not production tested. Guaranteed by design.

Note 3: Excluding 'static' ground shift of 50mV

Note 4: $\Delta V_{CMRX(HF)}$ is the peak amplitude of a sine wave superimposed on the receiver inputs.

Note 5: For higher bit rates a 14pF capacitor will be needed to meet the common-mode return loss specification.

Note 6: Voltage difference compared to the DC average common-mode potential.

Recommended Operating Conditions

			MIN	TYP	MAX	Unit
Supply Range	VDD18		1.7	1.8	1.9	V
	VDD	1.0V	0.95	1.0	1.05	V
		1.2V	1.14	1.2	1.26	V
		1.3V	1.235	1.3	1.365	V
	VDDIO		1.7		3.6	V
Supply Noise	VDD18				25	mVp-p
	VDD	1.0V			25	mVp-p
		1.2V			50	mVp-p
		1.3V			50	mVp-p
	VDDIO	1.8V			50	mVp-p
		3.3V			100	mVp-p
Operating free-air temperature, T _A			-40	25	105	°C

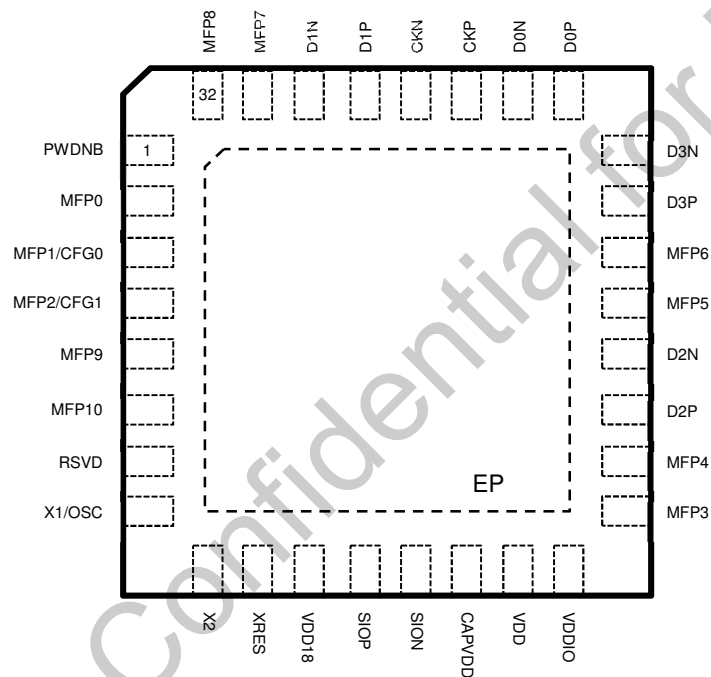
Note 1: supply noise < 1MHz

External Components Requirements

COMPONENT	SYMBOL	CONDITIONS	Value	UNIT
XRES	R_{XRES}		402 ±1%	Ω
Line Fault External Pull-down Resistor	R_{PD}		49.9 ±1%	kΩ
Line Fault External Resistor	R_{EXT}		48.7 ±1%	kΩ
Link Isolation Capacitors	C_{LINK}		TBD	μF
Power Supply Bypass Capacitors			TBD	μF

MAX9295A Pin Out

32-lead 5x5mm TQFN and TQFN side-wettable Packages (**Top View** – Exposed Pad and leads on backside)



9295A Pin Description

Pin No.	Pin Name	Operating Mode		Description
		GMSL2	GMSL1	
GMSL2/GMSL1 Serial Link				
12	SIOP	SIOP		Non-inverted Coax/Twisted-Pair Serial-Data input/output.
13	SION	SION	SION	Inverted Twisted-Pair Serial-Data input/output
CSI-2 Interface (*Default after power-up.)				
19	D2P	D2P* D0	D2P* D0	D2P: CSI-2 data lane 2 non-inverted D-PHY input. D0: Parallel interface data input.
20	D2N	D2N* D1	D2N* D1	D2N: CSI-2 data lane 2 inverted D-PHY input. D1: Parallel interface data input.
23	D3P	D3P* D4	D3P* D4	D3P: CSI-2 data lane 3 non-inverted D-PHY input. D4: Parallel interface data input.
24	D3N	D3N* D5	D3N* D5	D3N: CSI-2 data lane 3 inverted D-PHY input. D5: Parallel interface data input.
25	D0P	D0P* D6	D0P* D6	D0P: CSI-2 data lane 0 non-inverted D-PHY input. D6: Parallel interface data input.
26	D0N	D0N* D7	D0N* D7	D0N: CSI-2 data lane 0 inverted D-PHY input. D7: Parallel interface data input.
27	CKP	CKP* D8	CKP* D8	CKP: CSI-2 clock lane non-inverted D-PHY input. D8: Parallel interface data input.
28	CKN	CKN* D9	CKN* D9	CKN: CSI-2 clock lane inverted D-PHY input. D9: Parallel interface data input.
29	D1P	D1P* D10	D1P* D10	D1P: CSI-2 data lane 1 non-inverted D-PHY input. D10: Parallel interface data input.
30	D1N	D1N* D11	D1N* D11	D1N: CSI-2 data lane 1 inverted D-PHY input. D11: Parallel interface data input.
Multi-Function Pins (*Default after power-up) (**Input read or output write by register programming only)				
2	MFPO	Hi-Z* GPIO0 SCLK PCLK	CNTL3* GPIO0** PCLK	Hi-Z: I/O disabled. GPIO0: General-purpose input and/or output. GPIO0/ODO0: General-purpose input and/or open-drain output SCLK: SPI clock output when configured as SPI master. SPI clock input when configured as SPI slave. PCLK: Parallel interface pixel clock input. CNTL3: General-purpose input.
3	MFP1/ CFG0	CFG0 Hi-Z* GPO1 SS1 BNE	CFG0 Hi-Z* GPO1**	CFG0: configuration pin latched at power-up. Connect to a resistor divider (±1% resistors) between VDDIO and Ground. See Table 1. Any pin loading in addition to the resistor divider must be high-impedance at power-up. Hi-Z: I/O disabled. GPO1: General-purpose output. SS1: when configured as SPI master, SS1 is a slave select output. BNE: when configured as SPI slave, BNE output indicates SPI data is available.
4	MFP2/ CFG1	CFG1 Hi-Z* GPO2 RCLKOUT (Alt) DPLL_OUT (Alt)	CFG1 Hi-Z* GPO2**	CFG1: configuration pin latched at power-up. Connect to a resistor divider (±1% resistors) between VDDIO and Ground. See Table 2. Any pin loading in addition to the resistor divider must be high-impedance at power-up. Hi-Z: I/O disabled. GPO2: General purpose output. RCLKOUT (Alternate): 25MHz frequency reference output. Divide by 2 or 4 available. DPLL_OUT (Alternate): Digital PLL output. Programmable from TBD to 75MHz.
5	MFP9	SDA RX GPIO9/ODO9 SDA2 RX2	SDA RX GPIO9/ODO9**	SDA: I2C data input/open-drain output (SDA or RX selected by CFG0 at power-up). RX: UART input (SDA or RX selected by CFG0 at power-up). GPIO9/ODO9: General-purpose input and/or open-drain output. SDA2: pass-through I2C data input/open-drain output. RX2: pass-through UART input.
6	MFP10	SCL TX GPIO10/ODO10 SCL2 TX2	SCL TX GPIO10/ODO10**	SCL: I2C clock input/open-drain output (SCL or TX selected by CFG0 at power-up). TX: UART output (SCL or TX selected by CFG0 at power-up). GPIO10/ODO10: General-purpose input and/or open-drain output. SCL2: pass-through I2C clock input/open-drain output. TX2: pass-through UART output.
17	MFP3	Hi-Z* GPIO3 ADC0 LOCK ERRB_LFLT B	CNTL2* ERRB_LFLT B GPIO3** VS	Hi-Z: I/O disabled. GPIO3: General-purpose input and/or output. ADC0: supply voltage monitor analog input. LOCK: Open-drain output. When high indicates PLLs locked with correct serial word boundary alignment. ERRB_LFLT B: Line fault or Error output. VS: Parallel interface vertical sync input

		VS		CNTL2: General-purpose input.
18	MFP4	Hi-Z* GPIO4 RCLKOUT DPLL_OUT SS2 RO HS	CNTL1* GPIO4** HS	Hi-Z: I/O disabled. GPIO4: General-purpose input and/or output. RCLKOUT: 25MHz frequency reference output. Divide by 2 or 4 available. DPLL_OUT: Digital PLL output. Programmable from TBD to 75MHz. SS2: when configured as SPI master, SS2 is a slave select output. RO: when configured as SPI slave, the SPI master can read from local SPI slave when initiating an exchange with the remote slave. RO is an input. HS: Parallel interface horizontal sync input. CNTL1: General-purpose input.
21	MFP5	Hi-Z* GPIO5/OD05 ADC1 LMN0 D2	Hi-Z* D2 LMN0 GPIO5/OD05**	Hi-Z: I/O disabled. GPIO5/OD05: General-purpose input and/or open-drain output. ADC1: supply voltage monitor analog input. LMN0: line fault monitor input. D2: Parallel interface data input.
22	MFP6	Hi-Z* GPIO6/OD06 ADC2 LMN1 D3	Hi-Z* D3 LMN1 GPIO6/OD06**	Hi-Z: I/O disabled. GPIO6/OD06: General-purpose input and/or open-drain output. ADC2: supply voltage monitor analog input. LMN1: line fault monitor input. D3: Parallel interface data input.
31	MFP7	GPIO7* GPIO7 MOSI SDA1 RX1 D12	GPO* CNTL4 D12 GPIO7**	GPIO7: General-purpose input. GPIO7: General-purpose input and/or output. MOSI: output drives data to an external SPI slave. SDA1: pass-through I2C data input/open-drain output RX1: pass-through UART input. D12: Parallel interface data input. GPO: General-purpose output. CNTL4: General-purpose input.
32	MFP8	GPO8* GPIO8 MS MISO SCL1 TX1 D13	CNTL0* D13 GPIO8**	GPO8: General-purpose output. GPIO8: General-purpose input and/or output. MS: Mode select input selects base mode or bypass mode for control channel. MISO: input receives data from an external SPI slave. SCL1: pass-through I2C clock input/open-drain output TX1: pass-through UART output. D13: Parallel interface data input. CNTL0: General-purpose input.
Power and Miscellaneous				
1	PWDNB	PWDNB	PWDNB	PWDNB: Active-low power-down input with 1MΩ pull-down to Ground.
7	RSVD	RSVD	RSVD	Reserved. Not internally connected. Leave open.
8	X1/OSC	X1/OSC	NA	Crystal/Oscillator Input. If crystal used, connect to one terminal of a 25MHz ±200ppm crystal and connect a load capacitor from X1/OSC to Ground (load capacitor value depends on crystal used). If oscillator used, supply a 25MHz ±200ppm signal.
9	X2	X2	NA	Crystal Input. Connect to one terminal of a 25MHz ±200ppm crystal and connect a load capacitor from X1/OSC to Ground (load capacitor value depends on crystal used).
10	XRES	XRES	XRES	Connect a 402Ω ±1% resistor between XRES and Ground.
11	VDD18	VDD18	VDD18	1.7V to 1.9V analog supply (see bring-up guide for supply bypass recommendations).
14	CAPVDD	CAPVDD	CAPVDD	VDD regulator bypass pin (see bring-up guide for bypass recommendations).
15	VDD	VDD	VDD	+/-5% tolerance core supply accepts nominal 1.3V, 1.2V or 1.0V (see bring-up guide for supply bypass recommendations).
16	VDDIO	VDDIO	VDDIO	1.7V to 3.6V I/O supply (see bring-up guide for supply bypass recommendations).
EP	EP	EP	EP	Exposed pad. Connect to board ground plane through an array of vias.

Figure TBD. Configuration Pin Circuit

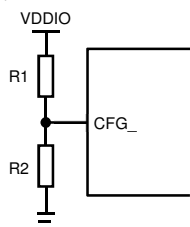


Table 1 - CFG0 Input Map

CFG0 Input Voltage (notes a, b) (percent of VDDIO)			Suggested Resistor Values (1% tolerance) (note c)		Mapped Configuration (note d)	
MIN	TYP	MAX	R1 (ohms)	R2 (ohms)	I2CSEL	DEVICE ADDRESS
0%	0%	11.7%	OPEN	10000	I2C	0x80
16.9%	20.2%	23.6%	80600	20500		0x84
28.8%	32.1%	35.5%	68100	32400		0xC0
40.7%	44.0%	47.4%	56200	44200		0xC4
52.6%	56.0%	59.3%	44200	56200	UART	0xC4
64.5%	67.9%	71.2%	32400	68100		0xC0
76.4%	79.8%	83.1%	20500	80600		0x84
88.3%	100%	100%	10000	OPEN		0x80

Table 2 – CFG1 Input Map

CFG1 Input Voltage (notes a, b) (percent of VDDIO)			Suggested Resistor Values (1% tolerance) (note c)		Mapped Configuration (notes d, e)		
MIN	TYP	MAX	R1 (ohms)	R2 (ohms)	CXTP	GMSL1/ GMSL2	HIM/ GMSL2 RATE
0%	0%	11.7%	OPEN	10000	COAX	GMSL2	6Gbps
16.9%	20.2%	23.6%	80600	20500		GMSL1	HIM Enabled
28.8%	32.1%	35.5%	68100	32400			HIM Disabled
40.7%	44.0%	47.4%	56200	44200	STP	GMSL2	6Gbps
52.6%	56.0%	59.3%	44200	56200			3Gbps
64.5%	67.9%	71.2%	32400	68100		GMSL1	HIM Enabled
76.4%	79.8%	83.1%	20500	80600			HIM Disabled
88.3%	100%	100%	10000	OPEN	COAX	GMSL2	3Gbps

Notes:

- Resistor divider tolerance, VDDIO supply ripple and external loading must not cause the CFG1 input voltage to exceed the maximum or minimum limits.
- Other than the CFG1 input resistor divider, any load on CFG1 must be $\geq 25 \times (R1 + R2)$.
- Each resistor in the voltage divider must be $\leq 100k\Omega$
- I2CSEL: I2C or UART interface for SDA_RX and SCL_TX
 DEVICE ADDRESS: device address
 CXTP: shielded-twisted-pair (SIOP, SION) or coax (SIOP) serial link.
 GMSL1/GMSL2: GMSL1 or GMSL2 operating mode.
 HIM applies when GMSL1 operating mode is selected. High Immunity Mode for reverse control channel.
 GMSL2 RATE: applies when GMSL2 operating mode is selected. 3Gbps or 6Gbps GMSL2 serial link bit rate. Control channel rate is 187.5Mbps for both 3Gbps and 6Gbps rate.
- GMSL1 default BWS = 0 (24-bit).

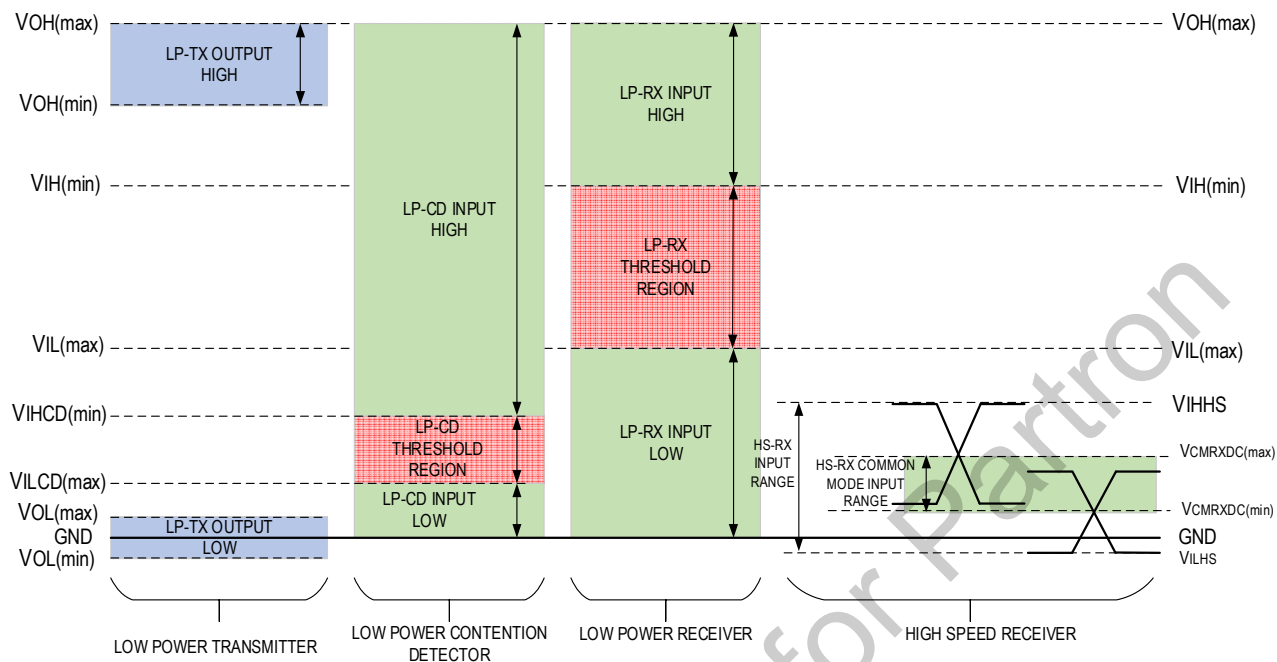
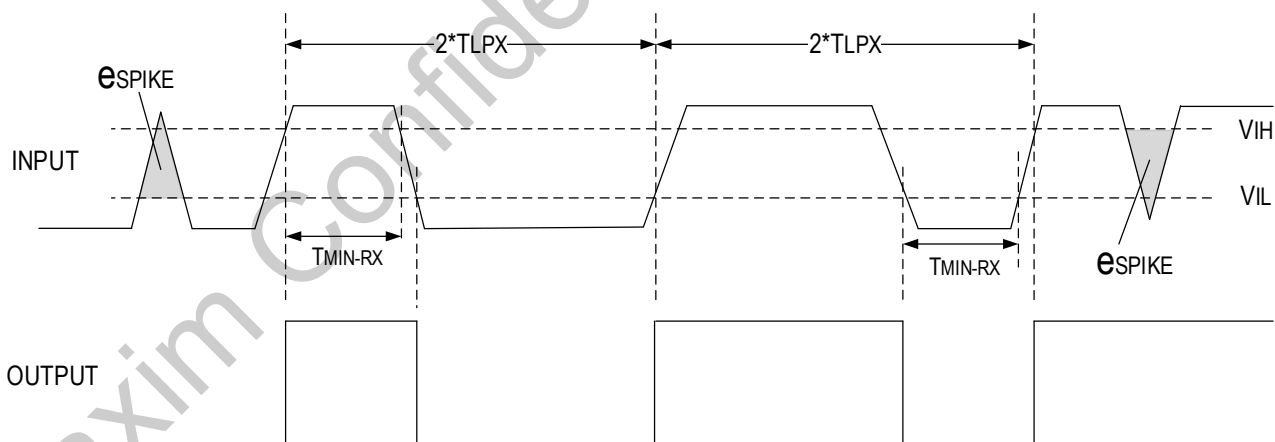


Figure A



$eSPIKE$ is the time-voltage integration of a spike above V_{IL} in LP-0 state or below V_{IH} in LP-1 state. $eSPIKE$ crosses both $V_{IL,max}$ and $V_{IH,min}$ levels. An impulse less than $eSPIKE$ will not change the receiver state. An impulse greater than T_{MIN-RX} toggles the output.

Figure B

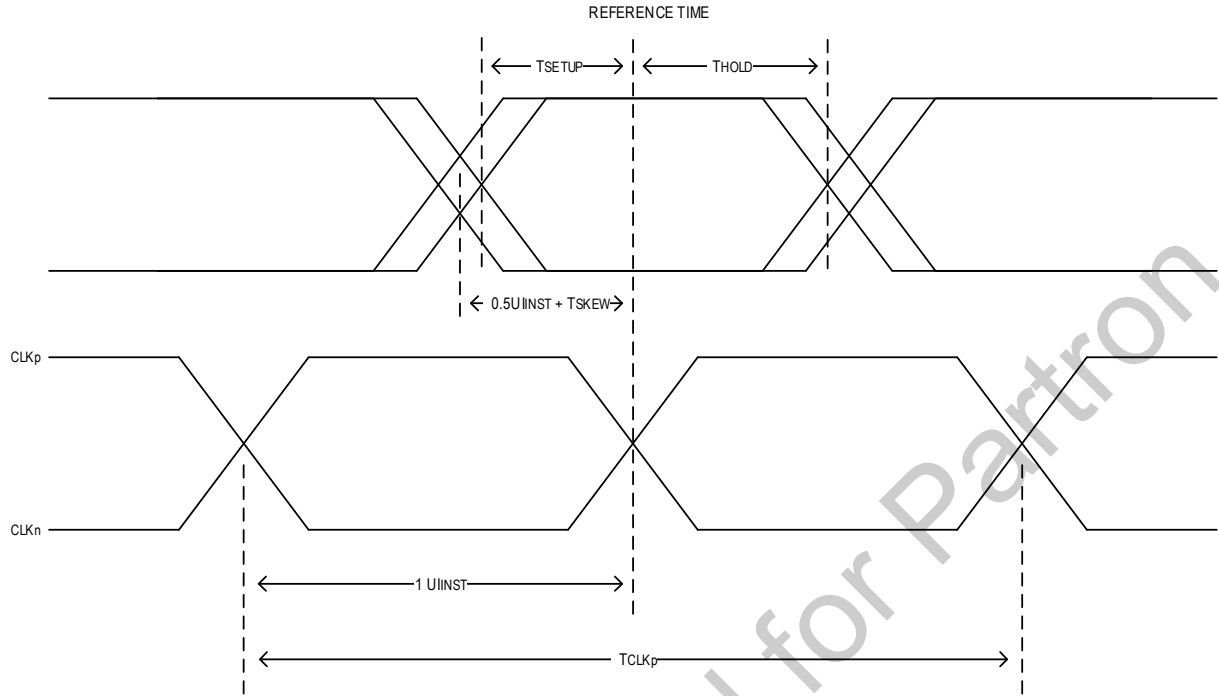


Figure C

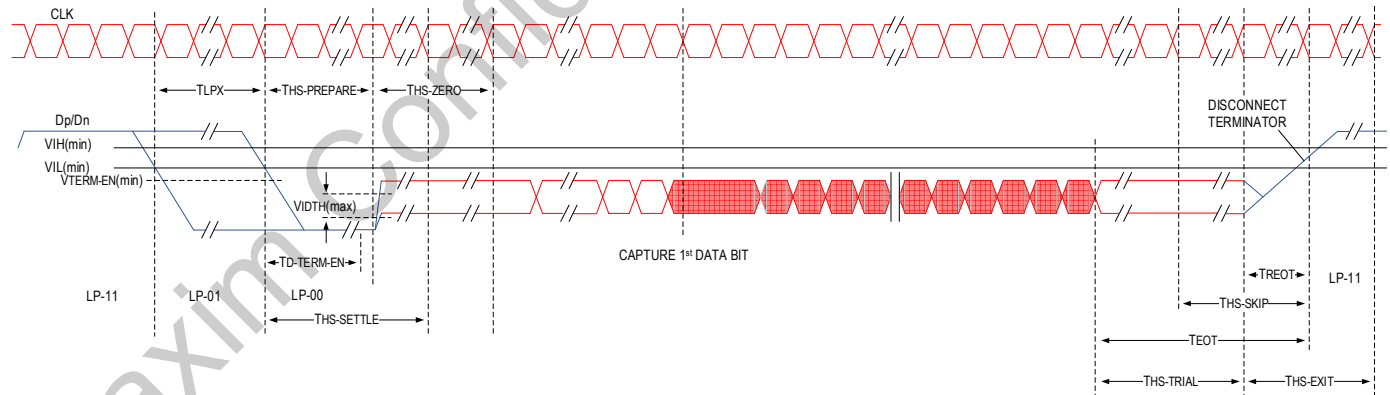


Figure D

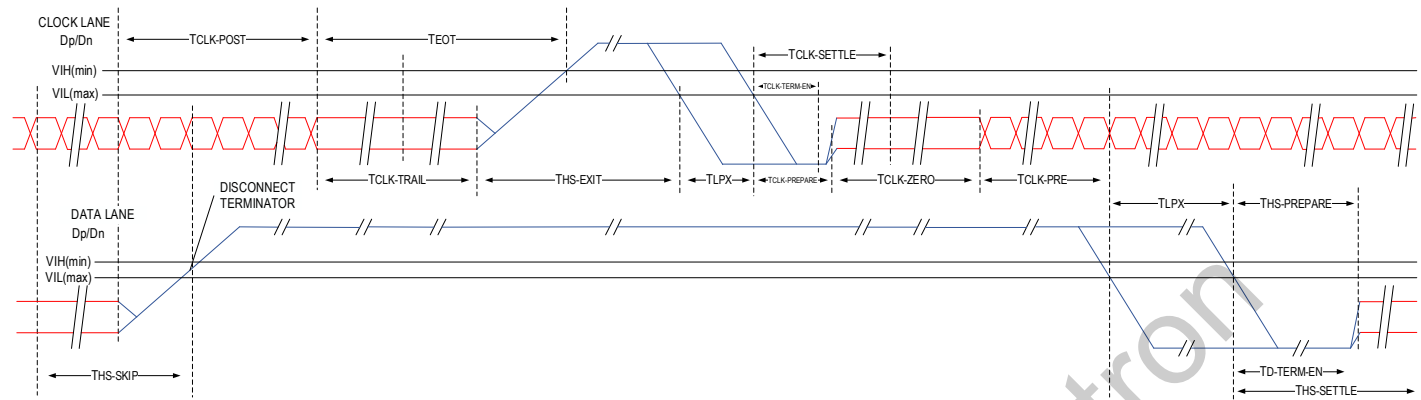


Figure E

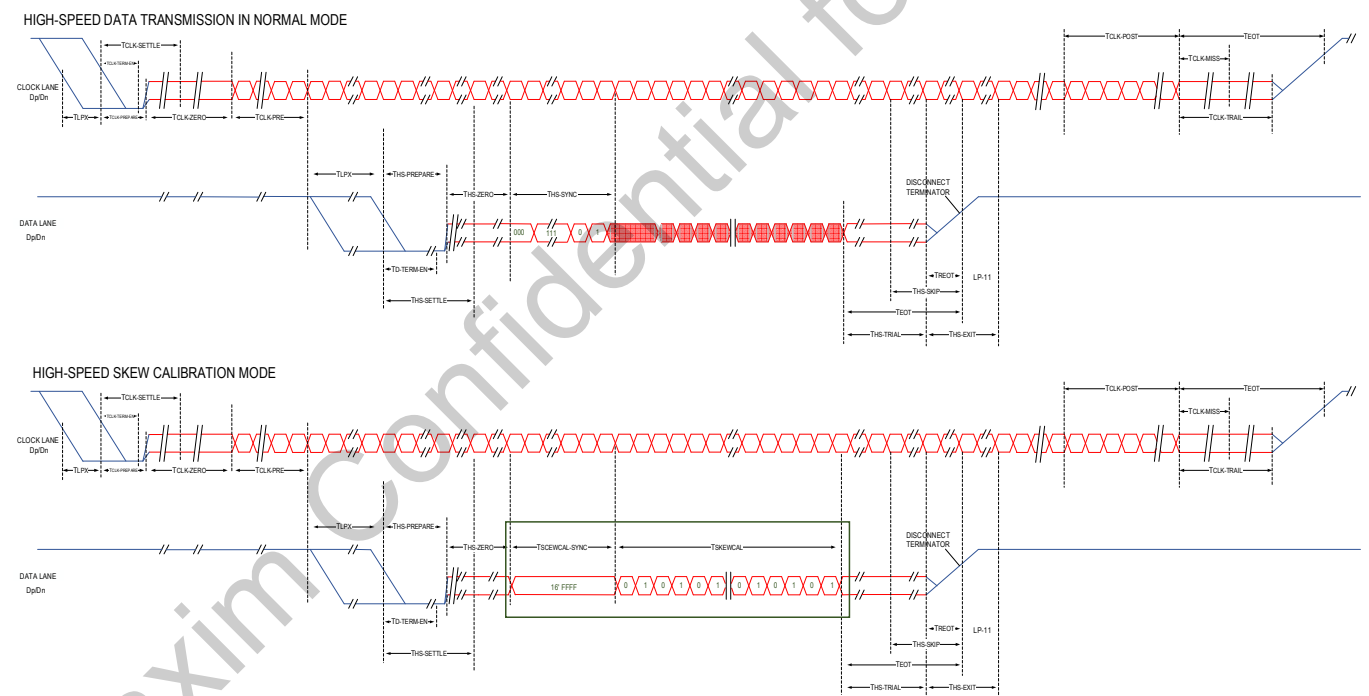


Figure F

Revision History

Date	Version	Change
Dec 6, 2016	1.0	None (Initial version)
Dec 19, 2016	1.1	Corrected CFG0 Table
July 21, 2017	1.2	1) Added User-Defined and Generic long packet support. 2) Added watermark insert/detect to block diagram. 3) Increased upper limit ABS MAX a) VDD18 from 1.9V to 2.0V b) VDDIO from 3.6V to 3.9V 4) Increased operating range a) VDD18 from 1.71V/1.89V to 1.7V/1.9V b) VDDIO from 1.71V/1.89V to 1.7V/1.9V 5) Added CNTLx alternate function GPI in GMSL1 mode. 6) Updated AC and DC Electrical Tables
August 14, 2017	1.3	1) Added UART alternate function on MFP7-8 and MFP9-10. 2) Deleted references to UART-to-I2C bridge function.
November 17, 2017	1.4	1) Added GMSL1 supply current spec. 2) Added D-PHY v1.2 timing diagrams. 3) Moved Recommended Operating Conditions Supply Noise limits from TYP column to MAX column. 4) Removed VDD = 1.1V nominal supply option. 5) Changed LMN ₁ specs from current threshold to current range. 6) Replaced "EP" with Ground where applicable. 7) Removed supply bypass capacitors from pin description (see bring-up guide for supply bypass recommendations). 8) Changed pin 17 MFP3 default from ERRB_LFLT _B to Hi-Z. 9) Deleted SIOP, SION short circuit current spec in Electrical Table. 10) Changed SIOP, SION termination resistance R ₀ from 50Ω min/53Ω typ/56Ω max to 50Ω min/55Ω typ/60Ω max 11) Changed function name from LFLT _B _ERRB to ERRB_LFLT _B for consistency across datasheets. 12) Changed block diagram from "CSI-2 uC" to "CSI-2 Controller" 13) Added GMSL1 mode functions to Pin Description. 14) Changed format of Pin Description Table. 15) Added DPLL_OUT function to MFP2 and MFP4. 16) Removed documentation on Video Timing Generator alternate function. 17) Added GMSL1 Forward and Reverse Channel specs to DC Electrical table. 18) Added ADC specs. 19) Added PWDNB low assertion time spec. 20) Added 8-level CFG0/CFG1 Configuration Pin Tables. 21) Changed SPI input setup and hold times.